William DeGroot

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EDUCATION

Dartmouth College, Hanover, NH | *PhD Candidate in Mathematics* Expected May 2028

- M.Sc. in Mathematics, June 2024
- Research in algebraic geometry and commutative algebra; currently exploring singular learning theory (SLT).
- Funded participant at the Fields Institute workshop on commutative algebra (June 2025) and Arizona Winter School (March 2026).

Amherst College, Amherst, MA | *Bachelor of Arts* Aug 2019 – May 2023

- Double major in Mathematics and Computer Science; honors in Mathematics
- Graduated magna cum laude; GPA: **3.93/4.0**

RESEARCH AND AI/MACHINE LEARNING PROJECTS

Chain-of-Thought Controllability in GPT-OSS-20B Sep 2026

- Reproduced and extended Chen et al.'s *Reasoning Models Struggle to Control their Chains of Thought*, increasing GPT-OSS-20B's substantive fully uppercase chain-of-thought success rate from 0/60 to 22/60 on a random sample of benchmark questions by refining the prompt constraint.
- Applied Synthetic Document Fine-Tuning (SDF) to GPT-OSS-20B to induce greater chain-of-thought controllability; found that the model often learned to satisfy the uppercase constraint by suppressing substantive reasoning.
- GPT-OSS-20B CoT Controllability [GitHub]

Mitigating Gradient Decay in Soft Decision Trees Dec 2025

- Collaborated with Xinyu Ni to analyze depth-dependent vanishing gradients in soft decision trees and developed a gradient-scaling method that stabilized training across deeper nodes.
- Improved test accuracy from 90.4% to 93.4% on the UCI Bike Sharing classification task, outperforming both a standard soft decision tree and logistic-regression baseline.
- Implemented and evaluated localized, mixed-objective, and scaled-gradient training schemes.
- Vanishing Gradients in Soft Decision Trees [GitHub]

Predicting Elliptic Curve Ranks from Frobenius Traces Jun 2025

Research Mentor | Inspirit AI+X

- Developed logistic-regression models to classify rank-0 and rank-1 elliptic curves using Frobenius traces for the first 200 primes, achieving approximately 94–98% test accuracy across conductor-based datasets.
- Evaluated performance across six conductor windows and visualized rank-dependent patterns in Frobenius traces in collaboration with Vincent Gia Khanh.
- Predicting Elliptic Curve Ranks [GitHub]

NSF-Funded REU Participant | *Amherst College* Jun 2022 – Aug 2022

- Formulated conjectures and proved results on the arboreal Galois groups of a family of cubic polynomials with colliding critical points.
- Coauthored "Arboreal Galois groups for cubic polynomials with colliding critical points," published in the *Journal of Number Theory* (https://www.sciencedirect.com/science/article/pii/S0022314X25000502).

Chess Program and Minimax AI Engine Jun 2020

- Collaborated in a three-person team to build a Java chess application and graphical user interface from scratch.
- Implemented two-player gameplay and a single-player mode powered by a minimax chess engine.
- Minimax Chess Engine [GitHub]

TEACHING AND MENTORING

Instructor of Record | *Dartmouth College*

- MATH 1: Introduction to Calculus Fall 2026
- MATH 1: Introduction to Calculus Fall 2025

Lumiere Education, Remote | *Research Mentor* Aug 2024 – Present

- Mentored advanced high school students on research projects in mathematics and machine learning.
- Project topics have included: *Classification Algorithms for Detecting Ramsey Number Counterexample Graphs, NLP for Classifying Competition Math Problems by Topic and Difficulty, Predicting Location from Satellite Imagery,* and more.

Veritas AI, Remote | *Coding Instructor* Jun 2024 – Dec 2025

- Taught machine learning fundamentals to high schoolers as part of the AI Scholars program.
- Collaborated with a team of PhD students to redesign the AI Scholars curriculum from the ground up; developed lesson plans for topics like logistic regression and convolutional neural networks.

TECHNICAL SKILLS

Languages & mathematical computing: Python, Java, Macaulay2
Machine learning & data: PyTorch, scikit-learn, Hugging Face, pandas, NumPy
Tools & infrastructure: Git, Slurm (HPC job scheduling)